

Project 2: Trading Signal Generator

Objective: Build a simple program that analyzes basic stock metrics and generates buy, sell, or hold signals based on user-defined thresholds.

Requirements:

1. Ask the user to input:
 - A stock symbol
 - Current price
 - 50-day moving average price
 - 200-day moving average price
 - Current RSI value (Relative Strength Index, a momentum indicator)
 - Current trading volume
 - Average trading volume
2. Define simple trading rules using comparison and logical operators:
 - Buy signal: Price is above both moving averages AND RSI is below 70 AND volume is above average
 - Sell signal: Price is below both moving averages OR RSI is above 70
 - Hold signal: Any other condition
3. Calculate and display:
 - Whether conditions for a Golden Cross (50-day MA crosses above 200-day MA) or Death Cross (50-day MA crosses below 200-day MA) exist
 - The percentage difference between current price and each moving average
 - The final recommendation (Buy, Sell, or Hold) with explanation of which conditions triggered the signal